Knowledge Integration in the Caffeine Literature

Where Useful Connections Between Research Communities Fail to Form: A Specification Outline

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Abstract. Caffeine is studied by pharmacologists, food chemists, sleep researchers, environmental scientists, analytical chemists, and cognitive psychologists. They share a molecule and very little else. That makes caffeine research a domain without being a discipline, and it raises a question that a more institutionally coherent field would not pose so sharply: when one of these communities produces knowledge another could use, does the knowledge actually travel? Researchers have named specific places where it should. Ricupero and Ritter, for instance, argue that caffeine's documented effects on vigilance and appraisal could be implemented in cognitive architectures, and that the caffeine literature would have to report its results differently for that to happen. This specification sets out how to test, against the citation record, whether such connections form. It distinguishes three things that are usually run together: whether two research communities address related problems, whether they draw on a common body of prior work, and whether they actually cite one another. Integration can fail at either transition, and the two failures mean different things.

Keywords: knowledge integration; citation networks; bibliographic coupling; community detection; Bayesian multilevel models; caffeine; cognitive science

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Status note, 13 October 2026 — start here when picking this work back up.

The construction defect is repaired; the community stage is unblocked. The era networks had keyed citing articles and cited references in two identifier spaces that never met, so no work appeared in both roles and no era network held a path of length two. The rebuild (§14.7) resolves both roles into the one canonical key space the 2023 pipeline used, dates every work by its own publication year, and writes the whole-corpus network alongside the era networks. It passes 23 checks, and 14 of the 17 complete eras now contain article-to-article citation. The analytic range is eras 11 to 17, 1971 to 2019.

What still needs doing. The CHAMP sweeps have not been rerun on the rebuilt networks, so the resolution question of §14.6 is open on better data rather than settled; C5 to C7 there remain. §14.6 describes the networks as they were before the rebuild and should be read as the record of a defect, not of the current construction.

What to carry into interpretation. The era networks hold 34.6% of the corpus's works and 24.6% of its citations [measured]. What falls outside is mostly recent, thinly cited work one era away rather than citation to canon, and descent is built from the full edge list, so those ties are not lost — but the communities themselves are induced from a quarter of the record. §14.7 gives the measurements.

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1 Purpose: Caffeine as a Test Case for Knowledge Integration

Why caffeine constitutes a research domain without being a field, and why that makes integration rather than fragmentation the interesting question.

1.1 Caffeine creates a domain, not a discipline

Caffeine supports many research agendas at once. Its pharmacological profile is unusually well characterized which makes it useful as a precise experimental instrument. Its consumption is nearly universal which gives findings about it relevance beyond research settings. The molecule also appears in enough contexts that very different research programs encounter it while pursuing very different questions. In the literature, caffeine is simultaneously a drug, a cognitive enhancer, a food constituent, an environmental contaminant, and an analytical target.

Caffeine research therefore constitutes a substantive domain without constituting a single discipline. The distinction is important. Researchers do specialize in caffeine, journals publish caffeine research repeatedly, and recognizable clusters of investigators know one another. What the domain lacks is a single institutional structure that encompasses it as a whole. The questions researchers ask about caffeine, the methods they use, and the literature they treat as foundational remain strongly shaped by their parent disciplines.

The result is a domain organized more like a watercooler than a library. Researchers arrive from different intellectual traditions, exchange knowledge, and move on, without any structure ensuring that what one group knows reaches another group that could use it.

The venue distribution reflects this structure. Of the 639,663 cited works whose journals can be identified from the Web of Science records, the ten most common journals account for 7.1% of the total, and the top fifty account for 19.9%. Reaching half requires 374 journals, drawn from 49,174 distinct venues [measured]. Across the full history of the corpus, the leading venues are broad biomedical and physiological journals: the *Journal of Biological Chemistry*, *PNAS*, the *Journal of Physiology*, *Nature*, and *Science*. Within the 2009–2020 window, however, the leading venues shift markedly toward food chemistry, environmental science, analytical chemistry, and applied biomedical research: *PLoS ONE*, *Food Chemistry*, *Science of the Total Environment*, the *Journal of Agricultural and Food Chemistry*, the *Journal of Chromatography A*, *Water Research*, and *Chemosphere* [measured]. A literature organized primarily around one or two research traditions would not be expected to show either this degree of venue dispersion or this much change in its leading outlets over time. The evidence does not show that caffeine research lacks institutions. It shows that no single disciplinary infrastructure organizes the domain as a whole.

> A note on what this establishes. The evidence shows that caffeine research is spread across venues belonging to substantively different domains. It does *not* establish that caffeine is unusually interdisciplinary, because that is a comparative claim and we have no matched benchmark. The argument only needs the weaker fact, so we assert only the weaker fact.

1.2 Which makes integration, not fragmentation, the question

Given all that, finding that the literature is fragmented would be unremarkable. We should expect distinct communities; the substance does not oblige a pharmacokineticist and an environmental engineer to read one another.

The interesting question is what happens when they *should*. Sometimes a finding produced in one corner of this domain is exactly what another corner needs, and the two corners have every substantive reason to connect. Does the knowledge move?

This is not a bibliometric abstraction we are imposing on the literature. Researchers have identified the connection points themselves. Ricupero and Ritter's (2024) review argues that caffeine's measured effects on vigilance, attention, and appraisal could be implemented directly in cognitive architectures such as ACT-R, which would let system designers predict how an operator's performance changes over a shift. They also identify why it has not happened: caffeine studies typically report statistical significance rather than effect sizes, doses rather than dosages, and no distributional information about half-life. The knowledge exists; it is not in a form the modeling community can use. Other reviews in this literature make structurally similar recommendations.

So the questions are: how often are proposed integration points actually realized, where has integration happened, and what do the successes and the misses have in common?

1.3 What we need to distinguish, and how bibliometrics lets us

Answering that requires separating three things that are easy to conflate:

1. whether two research communities **address related problems**;
2. whether they **draw on a common body of prior work**; and
3. whether they **actually cite and build on one another**.

Two long-standing bibliometric measures map onto the last two. **Bibliographic coupling** asks whether two bodies of work cite the same earlier sources. Two communities that are coupled draw on overlapping bodies of cited literature, whether or not either is aware of the other. **Descent** – direct citation from a later community to work belonging to an earlier one – is the stronger relation: a later community directly references work associated with the earlier one.

The first item, whether the problems are related, has to be measured *outside* the citation record. Communities are constructed from citation structure, so measuring their substantive similarity from that same structure would build the answer partly into the definition. The measure would not be empty – citation similarity does vary, and communities differ in density – but the inference would be circular in exactly the place where we need it not to be. We measure topical proximity instead from what the articles say: their abstracts, keywords, and publication venues.

Two further ideas matter later and are worth defining now. A **stochastic blockmodel** is a way of estimating how strongly groups connect to one another while accounting for the fact that large groups have more connections simply by being large. And a **generative model** is one that states how the observed counts could have arisen, which is what lets us say a particular connection is weaker than it should have been rather than merely small.

1.4 Caffeine as a case, and the general problem behind it

Interdisciplinary research is usually justified by the expectation that knowledge produced in one tradition will become useful in another. A great deal of institutional effort rests on that expectation, and the outcomes have often fallen short of it. Existing bibliometric work can count interdisciplinary publications and cross-disciplinary citations, which tells us where integration *occurred*. It does not tell us where substantively plausible integration *could have occurred and did not*.

That is the general problem, and caffeine is a case chosen because it offers unusually favorable conditions for observing it: one shared object, many parent disciplines, abundant literature, and – in Ricupero and Ritter and others – concrete prior claims that particular connections ought to exist. The method is designed to transfer to other interdisciplinary domains, with the caffeine case providing its first demonstration.

Finally, this is a literature that changes. Communities appear, grow, merge, and disappear. We will need to distinguish a *community* in a single year from a *literature* that persists across years, and from a *lineage* claim that one year's community is the same community as the next year's. Section 3 does that, and the distinction turns out to determine which questions we can answer with the data we have.

2 Two Views of the Literature Through Time

Two ways of building the networks, answering two different kinds of temporal questions.

The corpus is divided into eras – for the analytic window, single publication years from 2009 to 2020. How we assign works to eras determines what kind of temporal object we can study, and there are two defensible constructions that answer different questions. They are not competing implementations. They are different measurement instruments.

2.1 Path A: birth-era networks

A work belongs to the era in which it was published. Each era's network therefore contains that year's articles and the references entering the literature with them, and no work appears in two eras.

This description now holds, but it did not before October 2026. In `citation_node_list_30Oct2024.Rda` a cited work carries the year of the article citing it, because `Rank_Thinning_Function.R` joined the citing article's `time_id` onto both the source and the target column; the duplicate filter then kept whichever citing row appeared first in the file rather than the earliest year. The rebuilt construction (§14.7) dates every work by its own publication year, parsed from the CR string for cited works, and records the citing year separately as `year_first_cited`.

C1 resolved: the 2024 node list dated cited works by the citing article's year; the rebuild dates them by their own. §14.7

This construction supports questions about the *relationships between* communities in successive years:

- What knowledge does a later community inherit from an earlier one?
- Which communities draw on the same prior works?
- When does a shared body of knowledge translate into direct citation?
- Where do plausible connections fail to occur?

Descent and coupling live here, and so does the whole question of missed opportunities. This is the path the present analysis takes because the tools needed to construct the networks and identify their communities already exist.

2.2 Path B: active-era networks

A work belongs to *every* era in which it is active – published in that year or cited by an article from that year. Works therefore recur across eras, and because they recur, a community in one year can share members with a community in the next.

That shared membership is what makes it possible to follow a community through time, and so this construction supports a different class of question:

- Does a research community persist from one year to the next?

- Does it grow or contract?
- Does it split into several, or merge with others?
- Is it absorbed into a neighboring literature?
- What predicts how long it survives?

2.3 How the two relate

The difference is not implementation history. Path A's networks cannot answer Path B's questions because their communities share no members and therefore have no continuity to track; Path B's networks cannot sharpen Path A's descent relation because when works recur across eras, "cites something that was active in the previous era" becomes true of nearly every citation and stops distinguishing anything.

2.4 Which community partitions the final analysis uses

The partitions currently on disk were produced at different times by different procedures – eras 1–21 under one naming convention, eras 22–23 under another – and their community counts swing by a factor of 2.4 across three adjacent eras [measured]. That would be a problem for any temporal comparison if those were the partitions we intended to analyze.

They are not. The final analysis will generate every era's partition under a common procedure: a Julia implementation of CHAMP, which selects the community-detection resolution from the data rather than from a value someone typed, and applies the same selected resolution independently to each era. Because one procedure produces all of them, community counts become comparable across eras, and a change in the number of communities becomes a fact about the literature rather than an artifact of method.

The existing partitions remain useful as *development fixtures*: they let the orchestrator be built and tested against known outputs before the final partitions exist. They are not the partitions the inference will run on.

Pooled γ^*: not estimable on the old networks; the sweep has not been rerun on the rebuilt ones. §14.7

The plan described in the rest of this subsection is on hold. Two CHAMP sweeps over the integration window show that a pooled γ^* cannot be estimated from the Path A era networks by either of CHAMP's selection criteria, and that the size-invariance argument below cannot be tested on them. See §14.6.

One implementation detail of that plan has to be settled, and it is not a matter of taste. Under Path A a work belongs to exactly one era, so the era networks have disjoint vertex sets and an edge-pooled corpus network is block diagonal. The CHAMP penalty term is computed against the whole graph's mass, so era e in a pooled sweep sees an effective resolution of $\gamma\, m_e/m$ rather than γ — roughly a twelfth of the intended value with twelve eras of comparable size. A γ^* estimated on the pooled graph is therefore inflated by about m/m_e and does not transfer back to a single era. We estimate γ^* instead from the per-era sweeps and apply the pooled value to every era, which inverts the procedure of the source manuscript for a reason specific to this construction: there the periods shared members, so the pooled graph described a real object.

Path B is more than a refactor of the current pipeline: it requires rebuilding the networks, re-running community detection, and re-generating the community labels, and the memory demands are large enough that the construction is better done in Julia. But it is a second analysis of the same corpus, not a separate project. What Path B invalidates is the *reuse* of Path A's community partitions and their derived files for Path B's purposes – not Path A's results, which stand on their own networks.

The logical progression across the document runs from the narrowest relation outward: descent first, then coupling, then – in Path B – persistence.

3 Objects We Need to Distinguish

Four things that all get called "a research area" in ordinary speech, and why keeping them apart lets us answer a hard question with easy data.

Community detection gives us clusters within a single year. Everything else we might want to say about how research areas behave over time is a further claim, and the claims differ in how much they cost.

Object	What it is, and what claiming it commits us to
Article	An observable document: a caffeine paper, or a work one cites. No inference involved.
Community	A set of articles clustered together *within one era*. Under Path A, calling something "Community 17" in 2016 makes no claim whatever that there is a Community 17 in 2017. The label is local to the year.
Literature	A higher-order grouping of communities that occupy a common position in the citation structure. Several communities across different years may all belong to a pharmacokinetic literature without any one of them persisting intact. Identified by a nested stochastic blockmodel (§11), *not* by journal membership.
Lineage	The stronger claim that community A in 2016 *is* the predecessor of community B in 2017. This requires a rule for establishing identity through time, which Path A cannot supply.

The definition of a literature deserves one further sentence, because it is easy to assume we are simply grouping journals. We are not. Literatures are found from the citation structure itself, using a blockmodel that groups communities occupying similar positions in it. Journal composition and community labels then serve to *interpret* those groupings and to test whether they correspond to recognizable disciplinary boundaries. Discovery and validation stay separate, and the validation is meaningful only because the journal information played no part in the discovery.

The payoff is worth stating plainly, because it determines the shape of the project:

> We can study the rise and decline of a *literature* without solving the much harder problem of establishing *community* identity through time.

A literature's lifecycle is measured by how much of the field it occupies in each year – how many articles and how many communities belong to it, and what share of citation traffic passes through it. Those quantities can rise and fall without our ever claiming that any particular community survived. Community-level survival, which does require the identity claim, is Path B's business (§12).

4 Research Questions

Why a baseline has to come before any claim that a connection is missing.

4.1 Two temporal windows

The corpus spans 23 eras, from 1907–1944 through 2020. Those eras are not equal in width: the early ones cover decades, and eras 12–23 are the only unbroken run of single years. Two different spans are therefore in play, and confusing them makes several questions ambiguous.

Window	What it covers and what it is used for
Historical window	All 23 eras. Used to reconstruct the major literatures and describe how they emerge, grow, contract, and decline over the full history of the field.
Integration window	Eras 12–23, single years from 2009 to 2020, giving eleven adjacent era pairs on a constant one-year base. Used for the models of integration, where comparing across pairs requires that the pairs be commensurable.

Era width matters for the integration models because the descent relation looks one era back: a one-year window and a thirty-eight-year window select very different citations. It matters much less for describing a literature's trajectory, where the quantity is how much of the field the literature occupies and the eras are simply time points.

4.2 Why the questions come in tiers

We cannot call a connection "missing" merely because two communities fail to cite one another. Most communities probably should not cite one another – an environmental chemistry cluster and a sleep-deprivation cluster have little to say to each other, and their mutual silence is not a failure of anything.

So a claim about a missed opportunity requires three things in order. First, we need to know how integration *normally* works in this literature: given two communities with comparable overlap in what they read, how much citation ordinarily follows? Second, we need to know where those communities sit in the larger intellectual structure, because a shortfall between two clusters of the same literature means something different from a shortfall across a disciplinary boundary. Only then can we ask whether cognitive research departs from the pattern.

That is why the questions come in three tiers, and why the third is uninterpretable without the first.

4.3 Tier 1: what normal integration looks like

- **RQ1.** Do communities that work on related problems end up drawing on a common body of prior work, and how strongly?
- **RQ2.** Given a common body of prior work, how much direct citation ordinarily follows? *This is the baseline every later claim is read against.*
- **RQ3.** Are communities that draw heavily on the past also the ones the future draws on? That is, are being derivative and being generative related traits?
- **RQ4.** Does either conversion strengthen or weaken across the integration window?

4.4 Tier 2: the intellectual structure the communities sit in

- **RQ5.** What are the major literatures in caffeine research, and how do they emerge, grow, contract, and decline across the historical window – the full 23 eras, not only the recent ones?

- **RQ6.** Is the pharmacokinetic literature as large and persistent as the argument assumes? This tests the premise of the project rather than its conclusion – if that literature turns out to be small or transient, the claim that it offers underused knowledge weakens before cognitive science is examined.
- **RQ7.** Does the problems-to-shared-reading conversion differ among literatures? And do the communities the citation data produce line up with the disciplinary boundaries that journals imply, or cut across them?

4.5 Tier 3: whether cognitive research is different

- **RQ8.** Where does cognitive research sit – gathered into its own communities, or scattered through communities organized around other concerns?
- **RQ9.** Is the conversion from shared knowledge into citation *asymmetric* where cognitive research is involved? *This is the primary claim, and unlike the others it comes with a stated expectation.*
- **RQ10.** Which specific pairs of communities show a connection that should have formed and did not, and is cognitive research overrepresented among them?
- **RQ11.** Has the shortfall changed across the integration window?

4.6 What we expect to find, and why

RQ9 is not a neutral question about which side shows a shortfall. We expect a particular asymmetry, and stating it in advance is what makes the finding informative either way.

> **Expectation.** Conditional on shared knowledge and comparable opportunity, cognitive communities should be more likely to cite relevant neuroscientific and pharmacological caffeine research than those literatures are to incorporate higher-level cognitive constructs.

The reason is a difference in the level of abstraction at which the two literatures operate. Neuroscience and pharmacology characterize lower-level mechanisms – receptor binding, adenosine antagonism, coordination among neural systems. Cognitive science organizes such mechanisms under higher-order constructs: working memory, sustained attention, appraisal. A cognitive researcher has a standing reason to reach downward for mechanistic evidence, because the mechanism is what the construct is supposed to be implemented in. A neuroscientist studying receptor kinetics does not need to reach upward to a cognitive architecture in order to answer the question in front of them.

If that is right, the shortfall should appear on one side and not the other, and the direction is predicted rather than discovered. A symmetric shortfall, or one running the other way, would be evidence against this account of how the two fields relate.

4.7 One limit the corpus imposes on RQ9

The corpus consists of caffeine papers plus everything those papers cite. Work that never mentions caffeine is not in it at all. That asymmetry constrains what RQ9 can observe, and the constraint is worth making concrete.

Suppose cognitive caffeine researchers cite pharmacokinetic caffeine research. We can see that, because both sides are inside a caffeine-centered corpus.

Now suppose the reverse: pharmacokinetic researchers should be incorporating a general cognitive model – a theory of vigilance, say – that never mentions caffeine. That model is outside the frame entirely. Its absence from the corpus is guaranteed by how the corpus was built and tells us nothing about whether anyone should have cited it.

So absence cannot be read symmetrically. We can measure whether cognitive communities cite caffeine work, and whether caffeine communities cite cognitive work *that is already inside the caffeine literature*. We cannot measure whether they should have reached outside it. Any claim about direction is reported as a bound, not an estimate.

Sarah responded 22 September 2026; her answer is not yet written into the document.

Substantive decision 1: Whether the predicted asymmetry is theoretically right

The expectation above rests on a characterization of how cognitive science and neuroscience relate – that one organizes the other's mechanisms under higher-order constructs, and that the reaching-down relation is more natural than the reaching-up one. That is a claim about the two fields, not about the data, and Sarah is far better placed than the citation record to judge whether it is accurate. If the characterization is wrong the model still estimates both directions, but the paper would be reporting a discovery rather than testing a prediction, which is a weaker result.

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5 A Conceptual Model of Knowledge Integration

The argument as a diagram: what we think produces a citation between two communities, and which parts of that we can observe.

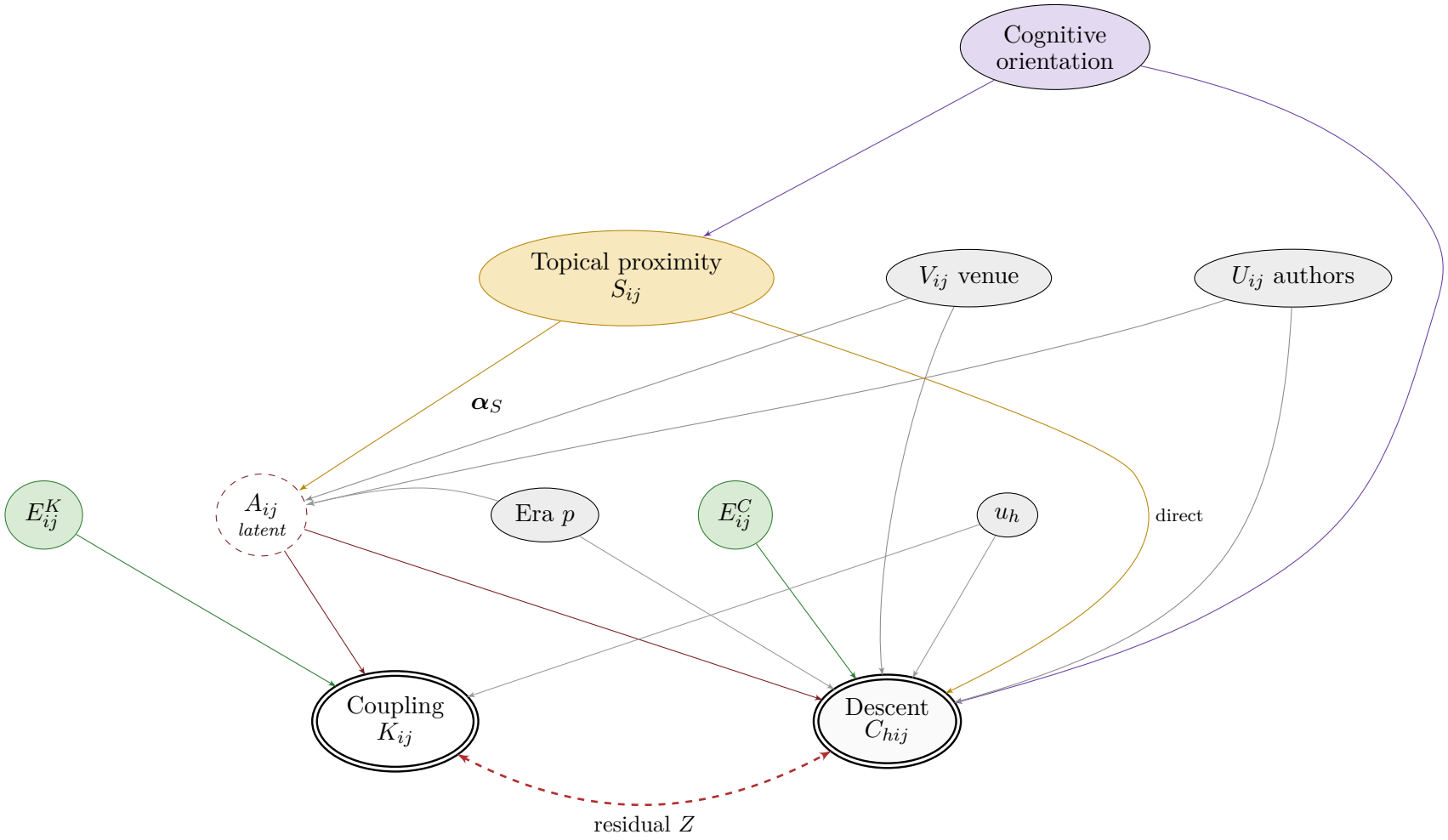


Figure 1: The conceptual model. Topical proximity S_{ij} shapes a shared knowledge base A_{ij}, which we cannot observe (dashed). Coupling K_{ij} and descent C_{hij} are two observed manifestations of it (double outlines). Remaining arrows are the alternative reasons a connection might form: shared authors U_{ij}, shared venues V_{ij}, opportunity E^K and E^C, individual work salience u_h, and era p. Drawn in Pearl's semi-Markovian notation; the dashed double-headed arc marks common causes of both indicators that we cannot measure, such as reciprocal reviewing obligations and shared institutional context.

5.1 Reading the causal diagram from left to right

The diagram organizes the argument around a latent **shared knowledge base** A_{ij} and its two observable manifestations, **coupling** K_{ij} and **descent** C_{hij}. The variables above and around them represent the conditions that can shape those relationships.

Start with **topical proximity** S_{ij}, shown in the upper-left portion of the diagram. Communities that study similar problems have more reason to encounter the same underlying work. We measure that proximity from what their articles say rather than from whom they cite. Topical proximity can affect integration in two ways. First, it can shape the shared knowledge base A_{ij}. Second, the diagram allows a direct relationship with descent, recognizing that communities working on similar problems may cite one another for reasons not fully captured by their shared references.

The **shared knowledge base** A_{ij} appears below topical proximity. It represents the overlap in what the two communities actually know about. We cannot observe that knowledge directly; nobody records everything researchers have read or know. The dashed outline therefore marks A_{ij} as latent.

At the bottom of the diagram are the two relationships we can observe. **Bibliographic coupling** K_{ij} provides one trace of the shared knowledge base. When two communities cite the same earlier works, they reveal common intellectual foundations. **Descent** C_{hij} provides a second, stronger form of integration: a later community directly cites work belonging to an earlier community. The later community is therefore not merely drawing on some of the same background literature; it is citing the earlier community's output directly.

The remaining variables represent other reasons these observed relationships may form. **Cognitive orientation**, shown at the top of the diagram, can shape both the problems a community studies and its propensity for direct citation. Shared authors U_{ij} and shared venues V_{ij} provide additional routes to connection. Opportunity E_{ij}^{K} and E_{ij}^{C} records how many chances the two observed relationships had to form. Work salience u_h allows some individual papers to attract citations across communities regardless of their substantive relationship, while era p accounts for temporal differences in the opportunity to accumulate citations.

Finally, the dashed double-headed arc between coupling and descent represents unmeasured common causes of both relationships. These may include shared institutional settings, reciprocal reviewing obligations, or other forms of intellectual contact that the citation record does not capture. The arc is important because it reinforces the central causal distinction developed in the next section: coupling and descent may covary without coupling itself causing descent.

5.2 Why coupling is evidence, not a cause

One feature of the diagram matters enough to state separately, because it changes what we are allowed to claim.

It is tempting to say that shared reading *causes* citation – that K leads to C. But bibliographic coupling is not something that happens to two communities before they cite each other. It is an observed consequence of their citation behavior, computed after the fact from the same reference lists. In the diagram, K and C are two manifestations of the same underlying thing, A. They covary because they share a cause, not because one produces the other.

> The shared knowledge base is a theoretical construct, not an estimated score. Coupling is *evidence* that it exists. We do not claim that coupling itself causes citation.

This has a practical consequence: A is not fitted. Two indicators are not enough to identify a latent variable without additional assumptions we have no grounds for, so we use K as an

observable stand-in and report the relationship between K and C as an association – how much citation accompanies a given degree of shared reading – rather than as a causal effect. Everything the argument needs survives that weakening.

5.3 Where the cognitive hypothesis lives

A causal diagram shows which variables affect which. It is not well suited to showing the claim this project actually makes, which is about the *strength* of a relationship rather than its existence. Figure 2 states that separately.

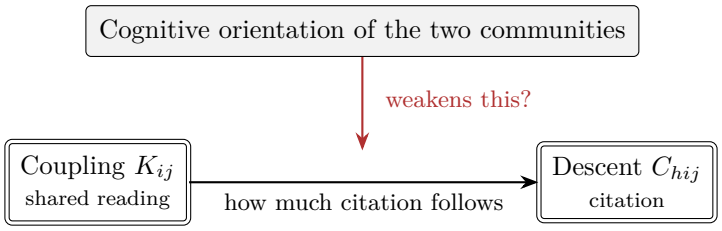


Figure 2: The claim under test is not that cognitively-oriented communities cite less in general. It is that the *conversion* of shared reading into citation is weaker where cognitive research is involved – they read the same works, and citation does not follow. Statistically this is an interaction, which is why it cannot be drawn as an arrow in Figure 1.

5.4 Two places where integration can fail

The chain has two transitions, and a shortfall at each means something different. This distinction is the analytical core of the project.

Stage 1 – from related problems to shared knowledge. If two communities work on similar problems but read largely different literatures, the barrier arises *before* any question of direct citation. They are not positioned to integrate because they do not share foundations. Nothing is being ignored; the two research programs have grown from different roots.

Stage 2 – from shared knowledge to direct citation. If two communities already draw on many of the same works and nevertheless do not cite one another, the situation is different. They have common intellectual foundations, and those foundations are not producing integration. This is the pattern that deserves to be called a missed opportunity, and it is what Ricupero and Ritter describe: a body of caffeine research whose findings the modeling community could use, built on overlapping foundations, yet not directly cited.

Naming these separately is what makes the finding interpretable either way it comes out.

5.5 Alternative explanations we have to rule out

Before attributing a shortfall to non-integration, four other mechanisms have to be considered. Each predicts something we can look for.

	Mechanism	What it predicts
M1	Vintage	Cognitive work in this corpus is more recent, so less of it has been available long enough to accumulate citations. If so, the shortfall shrinks once we account for how old the cited work is.
M2	Separation	High topical proximity, unexpectedly low coupling. The communities work on related problems and read different literatures. This is Stage 1 failing.
M3	Non-integration	High coupling, unexpectedly low citation. They hold foundations in common and citation does not follow. This is Stage 2 failing.
M4	Opportunity	Coupling and citation are exactly what the communities' sizes and activity levels predict. Nothing needs explaining. This is the null.
M5	Disciplinary distance	The shortfall tracks how far apart two communities are *disciplinarily*, not whether either is cognitive. If so, the cognitive result is an instance of a general norm rather than a fact about cognitive science.

M2 and M3 are the substantively interesting outcomes, and they are separated by *which* transition falls short – which is why the two stages are estimated separately rather than collapsed into one relationship. M4 is why Tier 1 has to come first. M5 is a genuine rival to the cognitive claim, and testing it requires disciplinary distance in the model alongside cognitive orientation.

These are mechanisms, not formal hypotheses. The formal questions are the RQs of §4; M1–M5 are the competing stories those questions have to adjudicate between.

6 Constructing the Networks and Measures

What exactly gets built: the nodes, the ties, and how each measure is computed.

6.1 The two network types

Three things are easy to run together here, so it is worth being explicit about the counting. There are **two network types** – descent and coupling – distinguished by what makes a tie. There are **three tie weightings** – breadth, fractional, and intensity – distinguished by what a tie's value counts. Each weighting applied to each type produces a different weighted network, so

$$2 \text{ network types} \times 3 \text{ weightings} = 6 \text{ weighted networks per era pair.}$$

All six share the same nodes. Within a type, all three weightings share the same ties as well, so one layout and one role partition serve all three.

For each adjacent pair of eras $(t{-}1, t)$, Path A produces two community-level networks over the same conceptual population.

The descent network.

- **Nodes.** Communities from the two eras. A community appearing in both roles is two nodes, one per era.
- **Tie.** An era-t community cites a work that belonged to an era-$(t{-}1)$ community and was actively cited during that earlier era.

- **Direction.** Directed, from the citing community to the cited one, following the citation.
- **Weight.** Yes – see §6.2.
- **Meaning.** Direct intellectual uptake. The later community reached into material that was part of the earlier community's own structure.

The coupling network.

- **Nodes.** The same communities.
- **Tie.** The two communities' articles cite at least one work in common, wherever that work sits.
- **Direction.** *Coupling is mathematically symmetric.* Sharing a reference is a property of the pair, not of one member. The Path A output orients it from the later era to the earlier one solely so that it aligns with the descent network and the two can be compared tie by tie; that orientation is a visualization convention and does not make coupling a directed relation. Because it is symmetric, a weight normalized within the later community and one normalized within the earlier community are both meaningful, and both are reported.
- **Weight.** Yes.
- **Meaning.** Common intellectual foundations, with no requirement that either community cites the other.

How restrictive descent is. Of the citations an era emits, 0.54% and 1.55% survive into the descent network on the two era pairs examined so far [measured]. That is the price of the strong definition: the cited work must both belong to the earlier era and have been in active use there. It is an indication of how narrow a one-year inheritance window is, and not a clean measurement of citation lag – it also reflects how the corpus was built and how the eras were partitioned, and separating those would take a decomposition we have not done.

6.2 Three ways to weight each network

Each tie can be weighted three ways, and each answers a different question. All three use the same nodes and the same ties; only the numbers change.

Weighting	What is counted	Question it answers
Breadth	Distinct shared works, each counted once for every community that cites it	How widely does one community draw on the other?
Fractional	Each shared work is one unit of evidence, divided equally among the community pairs it connects	How was one community's contribution distributed among its heirs?
Intensity	Citation events rather than distinct works; for coupling, the product of the number of citing articles on each side	How heavily does one community lean on the other?

Fractional weighting has one property the others lack: its totals do not change when the community partition is finer or coarser, because each shared work contributes exactly one unit however many pairs it spans. Breadth totals grow with the number of communities, and community counts vary by a factor of 2.4 across three adjacent eras [measured], so breadth totals are not comparable across pairs. Where a cross-year comparison is wanted, fractional is the weighting that supports it.

6.3 Topical proximity, measured outside the citation record

Topical proximity S_{ij} is how similar two communities' subject matter is, computed from the articles' abstracts, keywords, and journals – never from their references.

That independence is the whole point. Communities are *detected* from the citation network, so measuring their similarity from citations too would build the answer partly into the definition. Measuring similarity from text gives an independent yardstick, and only then does "these two communities are substantively close and do not connect" become a claim with content.

Naming the data source is not the same as specifying the measure, though, and this measure carries more weight than any other single quantity in the design: S_{ij} is what separates mechanism M2 from M3, and it is the covariate that makes a shortfall interpretable. It deserves to be built deliberately.

The problem a naive measure runs into. Every community in this corpus talks about caffeine. A conventional similarity measure over term-frequency vectors will therefore find every pair somewhat similar, because they share the domain's ubiquitous vocabulary – *caffeine*, *coffee*, *consumption*, *dose*. Worse, it rewards globally frequent terms, which are exactly the terms that carry no information about which community a document came from. If that happens, S_{ij} collapses toward a constant and stops doing any work.

A measure built on what distinguishes rather than what is shared. G. P. Morgan's corpora-comparison approach (2017) inverts the question. Rather than asking how much two bodies of text have in common, it asks, for each token: *if I saw this word, would I know which of the two corpora it came from?* Tokens are scored by a transformed normalized odds ratio, so a word that appears constantly in both contributes nothing, while a word concentrated in one contributes a great deal. The method was designed precisely to distinguish corpora by differential usage rather than by shared frequency.

Adapting it here is direct. Treat each community's article text as a corpus. For communities i and j, compute token-level polarization across the two, and define topical *distance* from the distribution of discriminating terms:

$$D_{ij}^{\text{text}} \;=\; f\big(\{\,\text{polarization}_{w,ij}\,\}_{w\in V}\big)\,, \qquad S_{ij} \;=\; 1 - D_{ij}^{\text{text}}$$

after normalization. The substantive reading is attractive: two communities are topically proximate when their vocabularies are *hard to tell apart*, not when both happen to contain the domain's boilerplate.

Morgan's procedure also suggests a preprocessing step we would otherwise have to invent. Its random-draw stage accumulates evidence about tokens that repeatedly fail to distinguish randomly constructed subcorpora – which is a principled way of identifying caffeine-domain boilerplate before any pairwise distance is computed.

A measurement study, not a commitment. We should not adopt this without checking it. Section 14 provides for a small measurement study comparing at least three candidates on the same community pairs:

1. corpora-comparison distance in the form above;
2. a conventional vector similarity over term frequencies, as the baseline the first is supposed to improve on;

3. an embedding-based similarity, if a suitable model is already available.

The validation is human. We select a set of community pairs whose substantive closeness we can judge from their labels and abstracts – some obviously related, some obviously not, some genuinely ambiguous – and ask which measure orders them the way a reader would. A measure that cannot separate the obvious cases is not going to be trusted on the ambiguous ones.

6.4 Cognitive orientation: two measures, deliberately kept apart

"Cognitive" can mean two different things about a piece of research, and in this corpus they come apart sharply.

- **Cognitive subject matter**, from abstracts and keywords: does the work study attention, vigilance, memory, executive function, reaction time?
- **Cognitive disciplinary provenance**, from journals: is the work published where cognitive scientists publish?

In era 23, 36.6% of caffeine articles carry cognitive subject matter [measured]. Meanwhile the four core cognitive-architecture venues – *Cognitive Science*, *Cognitive Systems Research*, *BICA*, and *Topics in Cognitive Science* – hold nine nodes between them in a corpus of 590,170 [measured].

A vigilance experiment published in *Psychopharmacology* is exactly the case that makes this worth keeping separate. Its content is cognitive; its provenance is pharmacological. Collapsing the two into a single "cognitive contribution" score would hide precisely the distinction that might explain why integration succeeds or fails.

We will write c_i for the cognitive orientation of community i – a proportion between 0 and 1, computed both ways, entering the model as two separate variables rather than an average.

A caution about the keyword measure: a draft lexicon produced false positives on `inhibit` (enzyme inhibition), `memory` (immunological memory), and `affect` used as a verb [measured]. The lexicon needs pruning and validation against a hand-coded sample before the measure is trusted, which is one reason the journal-based measure is worth having alongside it rather than instead of it.

6.5 Author and venue overlap

Two communities that share authors, or publish in the same journals, will cite one another more for reasons that have nothing to do with knowledge integration. Both are measured directly – author overlap from the author identifiers in the node list, venue overlap from the journal profiles – and both enter the model so that their contribution is not attributed to the process we care about.

Measurement decision 1: How topical proximity is computed

Three candidates are on the table (§6.3), and the choice matters because S_{ij} is what distinguishes "these communities were never going to connect" from "these communities should have connected". The empirical part is a comparison on the same pairs; the part that needs Sarah is the validation set. Judging which community pairs are substantively close – and which are close in a way that *should* produce citation – is exactly the expertise the citation record does not contain.

Sarah responded 22 September 2026; her answer is not yet written into the document.

Substantive decision 2: Which cognitive measure carries the primary claim

Subject matter and disciplinary provenance make different arguments. "Research on cognitive questions is not taken up" is a claim about what is studied; "work from the cognitive science community is not taken up" is a claim about who studies it. The model can carry both, but the theoretical argument should say which pattern we expect and treat the other as a competing explanation or a robustness check. Choosing after seeing the results would not be defensible.

7 From Networks to Statistical Inference

Why counting ties is not enough, and the four ideas that make the models intelligible.

7.1 The problem the networks cannot solve

The networks tell us which communities are connected and how strongly. They do not tell us whether an observed connection is stronger or weaker than it should have been.

Consider two community pairs, each connected by a single shared work. In one, the citing community published forty papers and emitted several thousand citations; in the other, it published two papers and emitted ninety. The same observed tie means something very different in the two cases. Likewise, some individual works are cited by nearly everyone regardless of discipline, so a tie running through such a work is weak evidence of anything.

Detecting a *missed* connection therefore requires a model of what citation would normally look like between communities with comparable opportunities. That is what the rest of this section builds toward.

7.2 Four ideas the models rest on

Exposure. How many chances did this community have to make this connection? A community that emits three thousand citations has three thousand opportunities to cite another literature; one that emits ninety has ninety. Exposure is a known quantity we compute from the data, and it enters the model as a fixed adjustment rather than something to be estimated – we are asserting that twice the opportunity means twice the expected count, all else equal.

Propensity. After accounting for opportunity, are some communities simply more inclined to draw on others, or to be drawn upon? This is not known in advance; it is estimated. Two communities with identical opportunity can still differ in how outward-facing they are, and that difference should not be mistaken for the phenomenon of interest. Operationally, propensity is not measured; it is a pair of varying effects, one attached to the citing community and one to the cited, that absorb whatever systematic inclination remains once opportunity and the covariates have been accounted for. A community that cites widely across the board raises its own effect rather than inflating the estimate for any one partner.

Work salience. Some individual works are cited far more than others for reasons particular to them – a canonical review, a widely used method. If we ignore this, a single famous paper shared between two communities inflates their apparent connection. Salience is likewise estimated rather than measured: it is a varying effect attached to the work itself, identified because most bridging

works are cited by more than one community, so the model can separate "this work is cited a great deal everywhere" from "this community cites this work unusually often." Section 9 makes both of these concrete – they are the terms θ_i, ψ_j, and u_h – and this is the reason the unit of analysis there is the individual work rather than the community pair.

Partial pooling. Many communities in this data are small. A community with one observed tie tells us very little on its own, and treating its observed rate as its true rate would be overconfident. A multilevel model handles this by estimating each community's parameter partly from its own data and partly from the field-wide pattern, with the balance determined by how much data that community actually has. Communities with a lot of data are barely shrunk toward the average; communities with almost none are shrunk a great deal.

This matters concretely here. Between 38% and 60% of participating communities have exactly one tie [measured], which under a simple proportion would give them a connection strength of 1.0 – not because their connection is exceptional but because arithmetic leaves no alternative. Partial pooling is what stops that from happening.

7.3 Two further ideas used later

The risk set. The model has to be told which community pairs *could* have connected. A pair where the earlier community holds no works eligible to be cited, or the later community emits no citations at all, is not a case of a connection failing to form; it is a case where no connection was possible. Those pairs are excluded. The remaining pairs – realized and unrealized alike – are the risk set, and the unrealized ones carry real information, because a connection that could have formed and did not is exactly what we are looking for.

Posterior prediction. These are Bayesian models, so what they produce is not a single estimate but a distribution over plausible values given the data. That gives us something a single number cannot: for a particular pair of communities, we can ask what citation count the model expects and how confident it is, and then ask how far below that expectation the observed count actually falls. Section 10 builds the definition of a forgone connection on exactly that.

8 Model I: From Topical Proximity to Common Knowledge

Stage 1. Do communities working on related problems end up reading the same things?

8.1 The question and the outcome

Model I asks whether topical proximity produces a common knowledge base, and how strongly.

The outcome is the number of works two communities cite in common – the coupling weight of §6. One row of data is one pair of communities in one era pair.

8.2 Why a negative binomial

The outcome is a count: it cannot be negative and has no upper bound in principle. The simplest count model, the Poisson, assumes the variance equals the mean. That assumption fails here, because community pairs differ enormously in how many chances they had to overlap, and because some pairs share a burst of works while most share none. The negative binomial is the same model with one extra parameter allowing the variance to exceed the mean, which is what this data does.

8.3 Exposure for coupling

Two communities can only share works they both had a chance to cite. If community i's articles cite R_i distinct works and j's cite R_j, and works were cited uniformly at random, the expected overlap would be proportional to $R_i R_j$. But citation is not uniform – a small number of works are cited enormously – so the expected overlap is larger than the raw product suggests. Writing d_w for the number of times work w is cited,

$$E^K_{ijp} \;=\; R_{ip}\, R_{jp}\, \kappa_p, \qquad \kappa_p \;=\; \frac{\sum_w d_w^2}{\left(\sum_w d_w\right)^2}, \tag{1}$$

where κ_p is a single number per era pair carrying the skew of the citation distribution. If every work were cited exactly once, κ_p would be $1/W$ for W works, recovering the uniform case.

8.4 The model

Writing K_{ijp} for the coupling count between communities i and j in era pair p:

$$K_{ijp} \sim \text{NegBinomial}(\lambda_{ijp},\, \phi_K) \tag{2}$$

$$\log \lambda_{ijp} = \underbrace{\alpha^K_p}_{\text{era baseline}} + \underbrace{\log E^K_{ijp}}_{\text{opportunity}} + \underbrace{\alpha_{S,p}\, S_{ijp}}_{\text{topical proximity}} + \underbrace{\alpha_{S,\ell}\, S_{ijp}}_{\text{by literature}} + \underbrace{\boldsymbol{z}^\top_{ijp}\boldsymbol{\gamma}}_{\text{other measures}} + \underbrace{\theta^K_i + \psi^K_j}_{\text{propensities}} \tag{3}$$

with the proximity slope allowed to drift across the eleven era pairs,

$$\alpha_{S,p} \sim \text{Normal}(\bar{\alpha}_S + \gamma_\alpha\,(t_p - \bar{t}),\; \sigma_\alpha)\,, \qquad \alpha_{S,\ell} \sim \text{Normal}(0,\; \sigma_\ell)\,. \tag{4}$$

8.5 What each term means, and which question it answers

Term	Meaning
$\bar{\alpha}_S$	How steeply shared reading rises with topical proximity, averaged over the eleven era pairs. **This answers RQ1.**
γ_α	Whether that relationship strengthens or weakens over time. **Half of RQ4.**
σ_ℓ	How much the relationship differs among literatures. If close to zero, the same rule holds everywhere; if large, some literatures convert proximity into shared reading far more readily than others. **Part of RQ7.**
θ^K_i, ψ^K_j	Each community's own tendency to share references, beyond what opportunity explains.
$\boldsymbol{z}$	Author overlap, venue overlap, disciplinary distance, and both cognitive measures for each community.
ϕ_K	The extra-variability parameter that makes this negative binomial rather than Poisson.

A community pair with high topical proximity and coupling far below what this model expects is mechanism M2: substantively related, reading different literatures.

9 Model II: From Common Knowledge to Citation Integration

Stage 2. Given that two communities read the same things, does citation follow – and is that weaker for cognitive research?

9.1 The question and the outcome

Model II asks how much direct citation accompanies a given degree of shared reading, and whether that conversion is weaker where cognitive research is involved.

The outcome is a citation count. But rather than counting citations between two communities in aggregate, we count them at the level of the individual work being cited.

9.2 Why the unit is the individual work

Some works are cited by almost everyone. If we aggregate to the community-pair level, such a work inflates every connection it touches and we cannot separate "these two communities are genuinely connected" from "they both happened to cite a canonical review". Estimating a salience term for each work fixes that – but a salience term needs a work to attach to, and an aggregated community-pair count has no single work in it. A pair connected by seventeen shared works would need seventeen salience terms.

So one row of data is one *bridging work* paired with one *citing community*: did era-t community i cite work h, and how often? The cited community j follows from h, since a bridging work belongs to a community in the earlier era by definition. For the 2019–2020 pair the risk set pairs 877 bridging works with every era-23 community that emits citations – up to all 394 of them – giving on the order of 345,000 rows, the overwhelming majority of them zeros [measured]. Only 151 of those communities appear in a realized tie; the rest contribute rows that are entirely zero, and those are precisely the observations RQ10 rests on.

Aggregating the fitted model back over the works belonging to a community recovers the community-pair quantity, so the question remains one about communities even though the data are finer.

9.3 Exposure for citation

At this resolution the opportunity is simply how many citations the citing community emits:

$$E^{C}_{hip} \;=\; N^{\text{out}}_{ip}. \tag{5}$$

A work's own attractiveness does not belong in the exposure term, because that is exactly what the salience parameter u_h is for.

9.4 Why the outcome is modeled in two stages

Citation between two communities involves two substantively distinct things.

First, does any citation relationship form at all? This is a yes-or-no question about whether the later community is in contact with this piece of the earlier community's output.

Second, given that some citation occurs, how much? This is a question about the strength of a relationship that already exists.

We model these separately, with one submodel for whether the count clears zero and a second for how large it is once it has. That separation is a theoretical commitment, not a convenience. The claim under test is that two communities read the same works and citation does not follow, which sounds at least as much like a failure of contact as a weakness of an established connection. Forcing both into one parameter would hide which it is, and the answer is substantively interesting either way: cognitive orientation might govern whether a connection forms at all while leaving established connections perfectly normal, or the reverse. Estimating the two stages separately lets each covariate carry a different coefficient in each – which is exactly the comparison RQ9 needs.

What this construction is called, and what it is not. A two-stage count model of this kind is a **hurdle model**. The name is unhelpful but the mechanics are simple: one process decides whether the outcome clears zero, and if it does, the size is drawn from a distribution that cannot produce a zero. Every zero in the data therefore comes from the first stage, and every positive count from the second.

The closely related alternative is a *zero-inflated* model, and the difference matters because the two are routinely conflated. In a zero-inflated model, zeros arise from two different sources: some communities are simply incapable of citing this work, and the rest cite it at some rate that happens to come out as zero. An observed zero is then ambiguous between the two, and the model has to apportion it. The hurdle form makes no such claim: a zero means one thing, that contact did not occur.

The hurdle form is the right one here because the risk set has already done the work that zero inflation would otherwise be doing. Pairs where citation was impossible – no eligible works, no emitted citations – are excluded before estimation (§7). What remains are pairs that could have connected, so a zero among them is informative rather than structural. That is precisely the case the hurdle form describes, and it is the case RQ10 depends on.

9.5 The model

For bridging work h belonging to earlier community j, cited by later community i:

$$C_{hip} \sim \text{Hurdle-NegBinomial}(\pi_{hip},\, \mu_{hip},\, \phi_C) \tag{6}$$

where π is the probability that any citation occurs and μ the expected count given that it does. Both parts take the same form:

$$\begin{aligned}\operatorname{logit} \pi_{hip} = {} & \eta_p + \log E^C_{hip} + \beta^{(H)}_{K,p} \log(K_{ijp}{+}1) + \beta^{(H)}_{Ki} \log(K_{ijp}{+}1)\, c_i + \beta^{(H)}_{Kj} \log(K_{ijp}{+}1)\, c_j \\ & + \boldsymbol{x}^\top_{hijp} \boldsymbol{\kappa} + u^{(H)}_h + \theta^{(H)}_i + \psi^{(H)}_j \end{aligned} \tag{7}$$

$$\begin{aligned}\log \mu_{hip} = {} & \alpha^C_p + \log E^C_{hip} + \beta^{(+)}_{K,p} \log(K_{ijp}{+}1) + \beta^{(+)}_{Ki} \log(K_{ijp}{+}1)\, c_i + \beta^{(+)}_{Kj} \log(K_{ijp}{+}1)\, c_j \\ & + \boldsymbol{x}^\top_{hijp} \boldsymbol{\delta} + u^{(+)}_h + \theta^{(+)}_i + \psi^{(+)}_j \end{aligned} \tag{8}$$

9.6 What each term means, and which question it answers

Term	Meaning
β_K	How much citation accompanies a given degree of shared reading, field-wide. **This is RQ2, the baseline everything else is read against.**
β_{Ki}	Whether that conversion is weaker when the *citing* community is cognitively oriented: cognitive communities not citing work despite shared foundations. **Part of RQ9.**
β_{Kj}	Whether it is weaker when the *cited* community is cognitively oriented: cognitive work not being cited despite common foundations. **Part of RQ9.**
u_h	How citable work h is in its own right, so a canonical paper does not manufacture apparent connections.
θ_i, ψ_j	Each community's tendency to cite and to be cited, beyond opportunity.
$\boldsymbol{x}$	Topical proximity, author and venue overlap, disciplinary distance, and the cited work's vintage – the last of these being the test of mechanism M1.

The two superscripts matter: $\beta^{(H)}$ governs whether a connection forms and $\beta^{(+)}$ how strong it is once formed, so RQ9 has two answers rather than one.

9.7 Change over time, and the two community traits

The baseline conversion and the cognitive terms both drift, so that RQ4 and RQ11 are estimated rather than inferred by comparing separate fits:

$$\beta^{(\cdot)}_{K,p} \sim \text{Normal}\Big(\bar{\beta}^{(\cdot)}_{K} + \gamma^{(\cdot)}_{\beta}(t_p - \bar{t}),\ \sigma_{\beta}\Big), \qquad \beta^{(\cdot)}_{K\cdot,p} \sim \text{Normal}\Big(\bar{\beta}^{(\cdot)}_{K\cdot} + \gamma^{(\cdot)}_{KC}(t_p - \bar{t}),\ \sigma_{KC}\Big). \tag{9}$$

RQ3 asks whether being derivative and being generative are related traits. A community in 2019 appears as a citing community in the 2018–2019 pair and as a cited community in the 2019–2020 pair, under the same partition – so we can ask about both of its tendencies without ever claiming it persists as the same community. They are given a joint prior:

$$\begin{pmatrix}\theta_c \\ \psi_c\end{pmatrix} \sim \text{MVNormal}\left[\begin{pmatrix}0\\0\end{pmatrix}, \begin{pmatrix}\sigma^2_{\theta} & \rho\,\sigma_{\theta}\sigma_{\psi} \\ \rho\,\sigma_{\theta}\sigma_{\psi} & \sigma^2_{\psi}\end{pmatrix}\right], \tag{10}$$

where ρ is the correlation RQ3 asks about. Model I supplies an analogous correlation for the tendency to share references, which is a different trait and is reported alongside rather than merged with it.

The model estimates both β_{Ki} (cognitive communities citing less than shared foundations predict) and β_{Kj} (cognitive work being cited less than shared foundations predict). These are different claims about how the field works, and the corpus limit of §4 bears on them differently. Which pattern we expect, and why, is stated in §4; it is registered there rather than here so that it is on record before the model is fit.

Sarah responded 22 September 2026; her answer is not yet written into the document.

Substantive decision 3: Whether Model I should also move to work resolution

Model II sits at work resolution because it needs a salience term. Coupling has the same dependence – one work shared by many communities generates related observations – but its work-level form needs work–community–community triples rather than work–community pairs, and we have not counted how many rows that would be. Until we do, Model I is specified at the community-pair level with the dependence acknowledged rather than modeled. Counting the triples is a half-hour job and should be done before the detailed specification.

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10 Identifying Forgone Connections

Turning "this connection should have happened" into two stated quantities: how far short, and how sure.

RQ10 asks which particular community pairs show a connection that should have formed and did not. Answering it requires a definition that does not depend on how we happen to sort a list, and that can be argued with – a reader should be able to disagree with a pair's placement on the list by disagreeing with a number, not with our taste.

10.1 What this is not

Because the language of "should have happened" invites the comparison, it is worth saying plainly what this procedure is not. It is not propensity-score matching, and it is not any other matched-contrast design.

Matching answers a question of the form: this unit received a treatment; what would have happened to a comparable unit that did not? It requires a treatment that units either receive or do not, and it constructs a control group of units with similar propensity to be treated, so that the difference in outcomes can be read as an effect.

Nothing here has that shape. There is no treatment: cognitive orientation, topical proximity, and shared reading are traits of communities, not interventions assigned to them. There is no control group: we are not comparing pair (i, j) to some matched pair (i', j'). And the counterfactual is not another unit's realized outcome; it is a distribution the model generates for *this* pair, given its own opportunity, its own partners' propensities, and the salience of the works involved. The comparison is within-pair and model-based, between what was observed and what the model considered plausible.

The word "propensity" does appear in this document, but in an unrelated and more literal sense: the varying effects θ_i and ψ_j, which capture how outward-facing a community is. They are parameters of the outcome model, not scores used to build a matched sample.

10.2 Why the obvious approach fails

The natural first move is to rank pairs by how far the observed count falls below the model's expectation. That does not work, for two reasons.

The gap is measured on the count scale, so it is mechanically larger for pairs with large expected counts. A pair expected to produce forty citations and producing thirty has a gap of ten; a pair expected to produce three and producing zero has a gap of three, and is far more interesting.

The gap also says nothing about confidence. A pair whose expectation is 10 ± 9 and a pair whose expectation is 10 ± 1 would rank identically at an observed count of two, though the second is strong evidence and the first is almost none.

10.3 Two quantities, doing two different jobs

The Bayesian machinery gives us a distribution of plausible counts for each pair, and that distribution supports both questions directly.

How far short. The expected shortfall is the gap between what the model predicts and what occurred:

$$D_{ijp} \;=\; \mathbb{E}\Big[\tilde{C}_{ijp} \;\Big|\; \text{data}\Big] \;-\; C_{ijp}. \tag{11}$$

This is a count, on the same scale as citations, and it answers "how much integration is missing." It is the quantity to report when the question is magnitude, and the quantity to sum over pairs when the question is how much integration the field forgoes in total.

How sure. The confidence that a shortfall is real, and materially large, is the posterior probability that the observed count falls below the prediction by more than a stated margin Δ:

$$F_{ijp} \;=\; \Pr\Big(\tilde{C}_{ijp} - C_{ijp} > \Delta \;\Big|\; \text{data}\Big). \tag{12}$$

F is a probability between 0 and 1. It is unitless, comparable across pairs of very different sizes, and it is the quantity to rank on.

The two are not redundant. D can be large while F is low, when the model's prediction is uncertain – a pair we cannot say much about. F can be high while D is small, when the model is confident about a modest shortfall – a real but minor gap. A pair worth writing about has both.

10.4 A worked example

Consider a citing community i in era p that publishes 30 articles and emits 2,400 citations, and an earlier community j holding 45 distinct works. The two share $K_{ij} = 22$ commonly cited references, which places them in the upper decile of shared reading for that era pair, and their topical proximity S_{ij} is high. One is oriented toward cognitive subject matter and the other is not.

Given that opportunity, that amount of shared reading, and the estimated propensities of both communities, the model's posterior predictive distribution for citations from i to j's works centers on 14, with an 80% interval running from 6 to 24. The observed count is 1.

Then $D_{ij} = 14 - 1 = 13$: thirteen citations' worth of integration did not happen. And with $\Delta = 3$, $F_{ij} = \Pr(\tilde{C} - 1 > 3) = \Pr(\tilde{C} > 4)$, which given that predictive distribution is roughly 0.93. The pair clears both bars and goes on the list.

Now change one thing. Suppose the citing community publishes 2 articles and emits 90 citations. The predictive distribution now centers on 0.5 and its 80% interval runs from 0 to 2. An observed count of 0 gives $D_{ij} = 0.5$ and $F_{ij} \approx 0.02$. Nothing is missing here: the community had almost no opportunity, and silence is what the model expects. This is the case that a raw proportion or a threshold rule would get wrong, and it is why the exposure term and the posterior predictive distribution are doing real work rather than decorating the argument.

10.5 The reporting rule

A pair is reported as a forgone connection when four conditions hold together:

1. the two communities share substantial reading, K_{ijp} above a stated quantile of the era pair's distribution;
2. they are substantively proximate, S_{ij} above a stated quantile;
3. the shortfall is material, $D_{ijp} > \Delta$; and
4. the model is confident, F_{ijp} above a stated cutoff.

Conditions 1 and 2 are what make the pair a candidate for connection at all; conditions 3 and 4 are what make the absence of connection notable. Reporting D and F together, rather than a rank, lets a reader see both how much is missing and how sure we are.

From works to pairs. Model II estimates at the level of the individual bridging work, so $\tilde{C}_{ijp}$ is obtained by summing the posterior predictive draws over the works belonging to j – draw by draw, so that the resulting distribution carries the correlation the varying effects induce among works from the same community. Summing point estimates instead would understate the uncertainty, and F would come out too confident.

Two things the quantity does not claim. It is not a claim that citation *would* have occurred under some intervention; the association it departs from is not a causal effect, for the reasons in §5. And "forgone" means materially less than expected, not literally zero. Requiring zero would discard the many cases of a connection that exists but is far thinner than the shared foundations warrant – which, given how many ties in this network rest on a single work [measured], is likely the more common form the phenomenon takes.

Sarah responded 22 September 2026; her answer is not yet written into the document.

Substantive decision 4: The materiality margin and the probability threshold

Δ and the cutoff on F determine which pairs appear on the list, as do the two quantile cuts on K and S. All four should be set in advance and reported with the sensitivity analysis described in §14.8, showing how the list changes as they vary. Choosing them after seeing which pairs they produce would make RQ10 unfalsifiable. The substantive question for us is what counts as a material shortfall: is Δ a fixed number of citations, or a proportion of the predicted count?

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11 Identifying Major Literatures and Their Lifecycles

What a "literature" is operationally, how we find them, and how we follow them through time without claiming communities persist.

11.1 What a literature is, operationally

Everything in this section rests on one definition, so it comes first.

> **A literature is a higher-order block: a group of communities that connect to the rest of the field in the same way.**

Communities are groups of *articles*, found within a single era. A literature is a group of *communities*, and it may span eras. It is neither a journal nor a keyword or label – those enter later, and for different jobs.

The criterion is structural equivalence in the blockmodel sense, and the phrase "connect in the same way" is doing precise work. Two communities belong to the same block not because they are densely tied to each other, but because they have the same pattern of ties to everything else: they draw on the same kinds of earlier work and are drawn on by the same kinds of later work. Two pharmacokinetic communities in adjacent years may barely cite each other and still be structurally identical, because the rest of the field treats them interchangeably. A grouping rule based on density would separate them; a blockmodel does not. This is why we use one.

11.2 Discovery, description, and validation are three separate steps

Conflating these is the standard failure mode of community-labeling work, so we keep them apart deliberately and give each its own instrument.

Discovery is structural. The blocks are estimated from the citation network of communities, and from nothing else. A **nested degree-corrected stochastic blockmodel** estimates which groups connect to which, at several levels at once – communities within literatures, literatures within broader divisions. *Degree-corrected* means it accounts for large communities having more

connections by virtue of size rather than affinity, which matters here because community sizes are wildly uneven. The number of groups is chosen by the data rather than set by us, using a minimum-description-length criterion that balances how well the grouping fits against how complicated it is: a grouping that needs many blocks to explain a little more is rejected in favor of a simpler one.

The output is not a single partition but a distribution over them, so each community carries a probability of belonging to each literature. That uncertainty is worth keeping. A community sitting ambiguously between two literatures is a substantively interesting object – it may be exactly where integration is happening – and a method that assigned it definitively to one would conceal that.

Description is textual, and comes second. Once the blocks exist, we say what each one is about, from the titles, abstracts, and keywords of its member articles and from the language-model labels of its member communities. This step is interpretive and it is downstream: the text never influences which communities are grouped together. Naming a block "caffeine and cardiovascular risk" is a claim about what the block contains, and it can be wrong without the block itself being wrong.

Validation is external. Journals give an independent partition of the same communities, derived from publication venue rather than from citation. Comparing the two answers the second half of RQ7. If the citation-derived literatures line up with journal boundaries, then what organizes this field is disciplinary affiliation; if they cut across it, something else is doing the organizing – which is the more interesting result, and the one §1 leads us to expect.

We report agreement with the **adjusted Rand index**, which measures how much two partitions of the same objects agree, corrected for the agreement expected by chance alone. Because a community can publish in several journals and because the blockmodel returns soft assignments, the headline ARI is computed on hard assignments – each community to its modal block and its modal journal – and accompanied by a soft analogue computed across posterior draws, so that the reported agreement carries the same uncertainty the partition does. Because journal identity is not derived from the citation record at any point, the comparison is not circular. Two caveats belong in the reporting: ARI is sensitive to the number of groups, so the journal partition should be coarsened to a comparable granularity before comparing; and a low ARI is a substantive finding here rather than a failure of the method.

11.3 How this differs from the community detection in §2

Two partitioning procedures appear in this document at two different levels, and they should not be confused.

	CHAMP (§2)	**Nested SBM (here)**
Objects partitioned	Articles, within one era	Communities, across eras
Produces	Communities	Literatures
Criterion	Modularity at a resolution	Description length
Chooses its scale by	Sweeping γ for a stable range	Model selection, automatic
Run	Once per era, in Julia	Once over the whole community network

CHAMP settles a question modularity maximization cannot settle on its own – which resolution parameter to use – by identifying the range of γ over which the partition is stable, and we apply a pooled γ^* across eras so that community granularity is comparable from year to year. It operates

on articles. The nested blockmodel then takes those communities as its nodes. Nothing about CHAMP determines the literatures, and nothing about the blockmodel revisits the communities.

11.4 Why not a threshold

An alternative to a blockmodel is to keep only the strongest ties and see what clusters remain. We avoid that for a specific and measurable reason: in these networks, between 72% and 90% of ties rest on a single shared work [measured]. Any threshold either removes almost the entire network or almost nothing, so the choice of cutoff would determine the answer more than the data would. This is the general weakness of threshold methods, and it is unusually severe here. A model that accommodates weak ties with appropriately wide uncertainty is preferable to one that deletes them and then reports the survivors as though they were the whole picture.

11.5 Lifecycles without lineages

Once communities are assigned to literatures, a literature's trajectory is measured by how much of the field it occupies in each era: how many articles belong to it, how many communities, and what share of citation traffic passes through it. Rise, growth, contraction, and disappearance are movements in those quantities, and because block membership is probabilistic, each is reported with uncertainty rather than as a count.

This is where the distinction of §3 pays off. None of these measurements requires knowing that a particular community persisted from one era to the next. A pharmacokinetic literature can be growing while its constituent communities dissolve and reform each year, and we can observe the growth without adjudicating the identity claim – which is the claim Path A cannot support.

12 Path B: Community Lineages and Persistence

The questions that need active-era networks, and the analysis available without them.

Section 2 set out what active-era networks would allow: following a community through time, and asking what predicts its survival. The natural model is a **discrete-time hazard model** – for each community in each year, the probability that it ends that year, given that it has survived so far. Because eras here are single years, discrete time is the natural form. A *competing risks* version separates two different endings, dying and being absorbed into another community, which the lifecycle vocabulary requires and a single-outcome model cannot distinguish.

What that model needs is an identity rule: something that says this year's community is last year's. Path A cannot supply one. A rule based on citation ties would be built on the same single-work ties that make thresholding unreliable [measured], and an unresolved identity algorithm should not hold up the rest of the project.

There is, however, a persistence analysis available now, and it needs no identity rule at all. Individual *works* have observable lifespans: we can see the first year a work is cited and the last, directly from the edge list. Modeling how long works stay in circulation – and whether that depends on the venue, the vintage, or the cognitive orientation of the communities citing it – tests something in the review's argument directly. If pharmacokinetic works stay in circulation longer than cognitive ones, that is evidence about which knowledge the field retains.

So the division is:

- **Path A now:** structure, integration, the cognitive case, literature lifecycles, and the persistence of cited knowledge.

- **Path B next:** persistence and transformation of the communities themselves.

13 Design Matrices

Reference tables: which column implements which quantity.

The tables below are for looking things up while implementing. Every quantity in them has been introduced and motivated in the preceding sections; nothing new appears here.

They are included because the gap between a model written in equations and a model a computer can fit is a table – one row per observation, one column per quantity – and most of the errors that survive a careful specification are errors in building it. Two in particular. The first is a row that should not exist: a community pair that had no opportunity to connect, or a bridging work paired with a community that emitted no citations that era. Those rows do not carry information about a connection failing to form, and including them would inflate the number of zeros the first stage of Model II has to explain. The second is a row that should exist and does not: a pair that could have connected and did not. Those rows are exactly the evidence RQ10 rests on, and they are the ones a naive construction drops, because they appear nowhere in the edge list – they have to be generated from the risk set rather than read off the data. The pipeline tests in §14 check both directions.

The unit of analysis differs between the three tables, and the difference is substantive rather than clerical. Model II is at work resolution because that is what identifies the salience effect u_h (§9); Model I is at pair resolution because coupling is a property of a pair and has no work-level analogue; the persistence table is at work–era resolution because it is a survival model, and a survival model needs one row for each period a unit was at risk. Aggregating the Model II table over the works belonging to a community must reproduce the Model I tie counts exactly, which is one of the consistency tests.

Each column plays one of five roles. The distinction between *exposure* and *covariate* is the one to watch: an exposure enters with its coefficient fixed at one on the log scale, which asserts that twice the opportunity means twice the expected count, while a covariate's coefficient is estimated from the data. Putting a quantity in the wrong role changes what the model claims, not just how well it fits.

Role	Meaning
Index	Identifies the observational unit – which work, community, era, or pair this row refers to. Not used in estimation.
Outcome	The quantity being explained.
Exposure	Known opportunity for the outcome to occur; enters with its coefficient fixed rather than estimated.
Covariate	A measured characteristic used to explain variation in the outcome.
Grouping	An identifier used for partial pooling: rows sharing a value share a varying effect.

13.1 Model II: one row per bridging work × citing community

A row exists for every combination of a bridging work h and a citing community i in the risk set – that is, for every work belonging to an earlier community that a later community *could* have cited, whether or not it did. The outcome `y_cites` is zero on most rows, and those zeros are the observations RQ10 is about.

Three grouping columns appear because three varying effects do: `work_id` groups the salience effect u_h, `citing_community` groups the citing propensity θ_i, and `cited_community` groups the cited propensity ψ_j. Note that `cited_community` is determined by `work_id` rather than being an independent index – each work belongs to exactly one earlier community, by the attribution rule the pipeline must fix (§14). Both cognitive measures appear for both communities, giving four columns, because §6 keeps subject matter and disciplinary provenance apart deliberately; the model is fit with one pair and refit with the other.

Column	Role	Quantity
`pair_id`, `era_t`, `era_prior`	index	Era pair
`work_id` h	index, grouping	Bridging work; groups the salience effect u_h
`citing_community` i	index, grouping	Groups θ_i
`cited_community` j	index, grouping	Follows from h; groups ψ_j
`y_cites`	outcome	Citations from i's articles to h; zero where none occurred
`out_citations_i`	exposure	E^C, citations emitted by i
`n_coupling`	covariate	K_{ij}, works i and j cite in common
`topical_prox`	covariate	S_{ij}, from text and venue
`disc_distance`	covariate	Disciplinary distance from journal profiles
`author_overlap`, `venue_overlap`	covariate	U_{ij}, V_{ij}
`cog_subject_i`, `cog_subject_j`	covariate	c_i, c_j by subject matter
`cog_venue_i`, `cog_venue_j`	covariate	c_i, c_j by provenance
`work_vintage`	covariate	Era of h; tests mechanism M1
`pair_index`	covariate	1–11, carrying the time trends

13.2 Model I: one row per community pair

A row exists for every ordered community pair in the risk set for that era pair. The exposure is stored as its three components rather than as a single product, so that the degree correction κ_p can be recomputed or varied without rebuilding the frame, and so that the exposure test in §14 has something to check against.

`literature_i` and `literature_j` come from the blockmodel of §11, which means this table cannot be built until that has been fit – the only ordering dependency between the two models. Because block membership is probabilistic, the column holds the modal assignment for the primary fit, and the sensitivity analysis refits across posterior draws of the partition.

Column	Role	Quantity
pair_id, community_i, community_j	index, grouping	
n_coupling	outcome	Works cited in common
R_i, R_j, kappa_p	exposure	Components of E^K
topical_prox, disc_distance, author_overlap, venue_overlap, cog_*	covariate	As above
literature_i, literature_j	grouping	Block membership, for $\alpha_{S,\ell}$

13.3 Persistence of cited works

This is the Path A persistence analysis of §12, and its unit is a work–era: one row for each era in which a work was still at risk of ceasing to be cited, running from the era it was first cited until either it stops being cited or the corpus ends. Works still in circulation at the end are censored rather than treated as having survived indefinitely.

cog_share_citers is the share of citing communities in that era classified as cognitively oriented, and it is the column that carries the substantive question: whether the field retains pharmacokinetic knowledge longer than cognitive knowledge. It is measured per era rather than once per work, because a work's audience changes.

Column	Role	Quantity
work_id, era	index	Work and year at risk
event, censored	outcome	Ceased to be cited; still cited at 2020
cog_share_citers, journal, era_first_cited, n_communities	covariate	Orientation of citers, venue, vintage, breadth of uptake

14 Pipeline Requirements and Tests

What the code must produce, and what must hold if it is producing it correctly.

14.1 Order of operations

The stages run in this order, and the order is not arbitrary – each depends on the last:

1. Resolve the era files, normalizing three inconsistent directory layouts and two partition naming conventions.
2. Build the era maps, checking that each network's vertices and its partition agree in length rather than assuming it.
3. Build the tie layer: deduplicate the edge list, apply the intersection filter, and aggregate once into a table at work resolution. This table is both the source of the descriptive networks and the model frame for Model II, so it is a deliverable rather than an intermediate.
4. Determine which communities participate in any tie. This defines the labeling queue, and it is what makes the next stage affordable.
5. Run Pajek to compute within-community degrees, for participating communities only.
6. Generate community labels, as a parallel queue against the local language model.

7. Assemble: join labels, write the networks, partitions, arc tables, and model frames.

Steps 4 and 6 are in that order for a substantive reason. Labels are currently generated for every community in both eras, but only communities that participate in a tie ever reach the graph – which means 67.6% and 87.9% of the labeling work is discarded on the two era pairs examined [measured]. Computing the ties first removes most of the dominant cost.

14.2 New quantities the pipeline must produce

- Exposure: citations emitted per community, distinct references cited per community, and the pair-level concentration constant κ_p.
- The risk set, with unrealized pairs represented explicitly rather than omitted.
- Topical proximity between communities, from abstracts, keywords, and journals.
- **Journal extraction.** The article-information step currently pulls titles, abstracts, and keywords from the Web of Science records but not the source field. Journal is needed three times over – to establish disciplinary breadth, to measure provenance separately from subject matter, and to supply the external partition for RQ7. Neither route available in the node list is adequate: 83.7% of node labels are bare DOIs, and a DOI prefix names a publisher rather than a journal – prefix `10.1016` alone covers 25.7% of the corpus – while the remaining 16.3% carry abbreviated journal strings that split one journal across variants, giving 7,099 distinct tokens for 95,882 works [measured]. The Web of Science source field is already in the records and is the better instrument.
- Author overlap between communities.
- Both cognitive measures per community, and vintage per bridging work.

14.3 Defects to repair

- **Arbitrary attribution.** When a work is cited by several communities, the current code credits it to whichever one happens to sort first in the edge list. This deletes 15.9% of ties on one era pair and 56.2% on the next [measured]. Because the damage differs by pair, the pairs are not comparable, which is fatal for an analysis whose point is comparison across years.
- **Duplicated edges.** Each citation is stored between two and fourteen times, non-uniformly [measured], so the fix is row deduplication rather than dividing by two.
- **Prompts for small communities.** Communities with fewer than five members receive prompts padded with the literal string `"NA"`, affecting 11.1% of one era's communities and 6.1% of the next [measured]. Since labels are how we recognize what a literature is, this sits on the critical path.

14.4 Tests

The tests below are grouped by what they protect.

Inputs. Every era resolves to an existing network and partition file; vertex count equals partition length; each era's vertex set matches the node list exactly (verified for all twelve eras of the integration window, in both directions [measured]); after deduplication the edge list has no repeated sender–target pair and 1,238,336 rows remain [measured].

Determinism. Shuffling the order of the edge list leaves every output bit-identical. This is the regression test for the arbitrary-attribution defect and the most important test in the suite.

Consistency between resolutions. Aggregating the work-level model frame over the works belonging to a community reproduces the community-pair tie counts exactly. This is what guarantees the two resolutions describe the same object.

Conservation. Fractional weights sum, per era pair, to the number of distinct bridging works; weights normalized within a community sum to one.

Exposure. E^K equals $R_i R_j \kappa_p$ on a hand-built example with known reference lists; κ_p equals $1/W$ exactly when every work is cited once.

Independence of the proximity measure. Topical proximity and disciplinary distance are computed by a routine that cannot see the edge list, enforced by a test rather than by convention.

Prompts. No prompt contains the literal string `"NA"`; every participating community receives a label.

Known-answer fixtures. The 2019–2020 pair reproduces 561 ties over 877 bridging works, with 151 of 394 citing and 234 of 794 cited communities participating [measured]; the 2018–2019 pair reproduces 251 ties over 126 bridging works [measured]. A change to these numbers is a deliberate specification change and the fixture is updated in the same commit that causes it.

These fixtures are development fixtures, and they are expected to change once. Eras 1–21 carry partition files named `era<N>_Community.clu` and eras 22–23 carry `era<N>_testCommunity.clu`. That naming split is not, however, where the granularity actually breaks. Community counts run 474, 455, then *1,126* across eras 17, 18 and 19 [measured], and era 19 has fewer vertices than era 18. Size cannot produce that. The cause is connectivity rather than resolution: era 18's network has a giant component holding 74% of its vertices and era 19's largest holds 5.4% [measured], and community counts track component counts because modularity cannot merge across components. Checked directly, only 5 of era 19's 1,112 components are subdivided at all, so 1,107 of its 1,126 "communities" are connected components [measured]. That question is not left open: §2 settles it by regenerating every era's partition with CHAMP at a pooled γ^*, which makes community granularity comparable across eras by construction. The current partitions are what the Path A orchestrator is developed and tested against; the fixture numbers above are rebuilt from the CHAMP partitions before any substantive result is reported, and no temporal claim in this document should be made from the current ones.

14.5 Implementation Status

Recorded as of 22 September 2026, so that a reader can distinguish code that exists from intended work. Throughout the document, [measured] marks a value measured from this corpus during implementation, rather than one inherited from a source or asserted by argument. The findings in the following subsection rest almost entirely on such measurements.

Built and passing. The Path A foundation is written in R under `scripts/Path_A/`: Pajek readers that return values rather than writing to the global environment, era-file resolution across the three directory layouts, era maps, and the tie layer. Its suite passes 38 checks. The known-answer fixtures reproduce 561 ties over 877 bridging works for 2019–2020 and 251 over 126 for 2018–2019, with 151 of 394 citing and 234 of 794 cited communities participating. These results confirm that the rewritten layer encodes the same both-periods reading as the original. Shuffling the edge list leaves bridging works, work-resolution ties, and descent weights bit-identical, so the arbitrary-attribution defect of §14 is repaired and covered by a regression test.

The community-detection stage is written in Julia under `community_detection_tools/`: Pajek format I/O, the Leiden stack, and CHAMP, behind one wrapper module. The Pajek suite passes 182 checks, including every era of the integration window; the Leiden suite passes 33 and the CHAMP suite 67.

The scale-assessment stage is written and has been run over the whole integration window. `champ_era_sweep` sweeps one era and returns its hull coefficients, its selected resolution, and a row of diagnostics; `write_champ_sweep` and `write_champ_summary` record them; the driver `champ_sweep_eras.jl` runs eras 12 through 23. Two complete sweeps have been run, one over $\gamma \in [0.1, 1.8]$ and one over $\gamma \in [0.01, 1.0]$. Their results are the subject of §14.6, and they are the reason the community stage is now blocked on a construction question rather than on code.

Written, not yet run. Coupling and the exposure constants (§8), built on sparse incidence matrices so that all three weightings follow from one multiplication. The hand-built fixtures pass; the corpus implementation remains untested.

Not started. Topical proximity, journal extraction, the cognitive measures, the labeling lane, and the orchestrator itself.

14.6 Open Problem: The Era Networks Carry No Detectable Community Scale

Superseded in part. Everything below describes the networks built before October 2026, whose construction is now known to have been defective in two ways: citing articles and cited references were keyed in separate identifier spaces, and cited works were dated by the year they were cited rather than the year they were published. §14.7 records the rebuild and what it changed. The resolution findings here have not been re-measured on the rebuilt networks, so they stand as a description of a defect rather than of the current construction; the sweep machinery, the diagnostics C5 to C7, and the reasoning about what a flat modularity envelope means all carry over unchanged.

> ***The Path A era networks are almost partitioned before community detection runs. They consist largely of disconnected stars, each one article and the references it cites, because an arc is kept only when both of its endpoints fall inside the same era — which discards roughly 70% of the citation record. Two complete CHAMP sweeps show that neither of CHAMP's selection criteria finds anything in these graphs: the elbow tracks the lower edge of whatever γ grid it is given, and no partition holds the modularity envelope over a wide interval. The obstacle is the construction, not the algorithm, and it must be settled before the community stage can proceed.***

This subsection records what has been measured, states the construction question the measurements raise, and lists the diagnostic checks that would answer it. None of it is resolved. It is written so that the work can be resumed from a cold start.

BLOCKING. Rerun the CHAMP sweeps on the rebuilt era networks (§14.7), eras 11–17. The construction question is answered; the resolution question is not.

What an era network contains. Era 12 is the 2009 slice. It holds 18,256 vertices: 989 caffeine articles returned by the Web of Science search and 17,267 cited works [measured]. Of those vertices, 965 emit at least one within-era arc and 17,345 receive one, at a mean of 18.6 arcs per citing article [measured]. Of the works receiving a citation, 96.8% are cited exactly once [measured]. The graph falls into 638 weak components, 494 of which consist of a single citing article and the works it cites. In every one of those 494 components, each vertex other than the citing article has degree exactly one and emits nothing [measured]. Mean degree stays near 2.0 across the integration window, and the same pattern holds in later eras: era 23 has 1,823 emitting articles at a mean of 31.0 arcs each, with 92.4% of cited works cited once [measured].

Why the networks are this sparse. The immediate cause is the within-era rule. An arc survives into an era network only when both endpoints belong to that era, and the twelve era networks together hold roughly 368,000 arcs out of about 1,238,000 in the deduplicated edge list [measured]. Approximately 70% of the citation record is therefore invisible to community detection. Whatever else is decided, a construction that restores structure has to return those arcs to the graph that detection runs on.

The second cause is unresolved, and it matters for how §2 describes Path A. The node list does not appear to date cited works by their publication year. The work labelled `critserjk_1988_fertilsterilNA` carries year 2009 and era 12; so does the work with DOI `10.1016/S0197-4580(97)00002-X` [measured]. Every one of the 590,170 labels occurs exactly once, and no cited work appears in more than one era [measured]. The natural reading is that a cited work receives the year in which it was *first cited* by an article in the corpus, not the year it was published. The arithmetic supports that reading: if arcs were same-year citations in the publication sense, a citing article would have on the order of one within-era arc, not the 18.6 to 31.0 observed. Under first-citation dating, an era network is the year's articles together with the references none of them had cited before, which is a star structure almost by construction.

This has not been confirmed against the edge list, and it should be (check C1 below). An earlier version of this subsection attributed the sparsity to the rarity of same-year citation; that explanation is inconsistent with the observed out-degrees and has been withdrawn.

What CHAMP does on these networks. Both sweeps were run directed and unweighted, with weak components of fewer than four vertices dropped beforehand. That filter removes very little, because the stars themselves average about twenty vertices: era 12 keeps 18,110 of 18,256 vertices and 567 of 638 components [measured].

Era	Giant share	γ^*	Communities	Components	Widest interval
12	6.1%	0.035	568	567	0.19
13	18.4%	0.061	565	560	0.14
14	11.4%	0.061	560	555	0.23
15	17.7%	0.035	534	530	0.13
16	51.8%	0.035	462	449	0.22
17	72.1%	0.035	341	330	0.25
18	74.1%	0.035	310	297	0.26
19	5.4%	0.035	948	947	0.29
20	4.8%	0.137	908	904	0.23
21	36.2%	0.035	797	786	0.32
22	58.1%	0.035	620	608	0.13
23	88.3%	0.035	229	219	0.18

All values are measured [measured]. Giant share is the largest weak component over all vertices; γ^*, communities, and components come from the sweep over $\gamma \in [0.01, 1.0]$ with 40 grid points; the widest interval is the longest stretch of γ over which any single partition holds the upper envelope, computed from the recorded hull coefficients.

Three readings follow.

The elbow tracks the grid floor. In the first sweep, over $[0.1, 1.8]$, seven of twelve eras selected 0.159, the second point of that grid. In the second sweep, over $[0.01, 1.0]$, nine of twelve selected 0.035, the second point of *that* grid, and two more selected the third [measured]. The selected value fell by roughly the ratio of the two floors. Era 23 moved from 0.628 to 0.035. Widening the window did not reveal an elbow that the first grid had cut off; it relocated the answer. The reason is structural: below the first split of the giant component, modularity prefers one community per component, so the sharpest change of slope on the envelope is always at the bottom of whatever range is swept.

No partition is stable across a wide interval. CHAMP's other standard criterion is the length of the γ interval over which one partition stays on the upper envelope. The widest such interval in any era is 0.32 out of a swept range of 0.99, and most lie between 0.13 and 0.25 [measured]. Between 13 and 22 distinct partitions take turns holding the envelope in each era. In four eras the widest interval runs up against the ceiling at $\gamma = 1.0$. The elbow criterion is pulled to the floor and the width criterion drifts to the ceiling; neither is locating a property of the network.

The selected partition is the component partition. At the selected resolutions, every era returns its components plus a handful of splits: era 12 gives 568 communities against 567 kept components, era 23 gives 229 against 219 [measured]. The gain in modularity over simply labelling each component ranges from 7.5×10^{-6} in era 12 to 0.014 in era 23 [measured]. Modularity itself exceeds 0.985 in every era, for a reason unrelated to mesoscopic organization: when communities coincide with components, nearly every arc is internal and the null-model penalty stays small, so Q approaches one arithmetically. That gain figure is also resolution-dependent rather than a property of the era. Era 23 showed a gain of 0.47 at the first sweep's $\gamma^* = 0.628$ and 0.014 at the second sweep's $\gamma^* = 0.035$ [measured].

The one durable result is an ordering. Eras 23, 18, 17, and 22 show the most structure and eras 19, 20, and 12 the least, in both sweeps, and that ordering follows giant-component share.

What this changes in the rest of the document. Three things.

First, the pooled γ^* described in §2 cannot be estimated from these networks by either criterion. It is not a matter of choosing between per-era and pooled estimation; there is no fixed point for either to find.

Second, the size-invariance argument of §2 cannot be tested on them either. That argument holds only under a condition these networks fail:

> Stability of γ^* across eras is evidence for a common mesoscopic scale *only when each era contains enough connected structure for the modularity envelope to discriminate among plausible partitions*. Without that condition, invariance in γ^* is not evidence of scale invariance at all; it is evidence of flatness.

Third, and most consequential, the communities themselves are in question. Every model in this document takes the community as its unit. If an era's partition is its components, then most communities are one article and the works it cites, and the objects the models are about are artifacts of the construction. The tie layer is unaffected — it reads the edge list and the era maps, not the era networks, so descent and coupling do not depend on this — but what those ties run *between* does.

Diagnostic checks, in the order they should be run. None of these has been done. C1 through C4 are estimable from the edge list and node list in about an hour and require no community detection.

1. **C1. The dating rule.** Join each cited work's recorded year to the earliest year among the articles citing it in the edge list, and count how often the two agree. Near-universal agreement confirms first-citation dating, which would mean §2 describes Path A incorrectly and that the both-periods reading of descent needs restating.
2. **C2. The citation-lag distribution.** How many years separate a citing article from the work it cites. This requires true publication years for cited works, which are not in the node list if C1 confirms first-citation dating; they can be parsed from the labels of the non-DOI entries, which carry the form `author_year_journal`. The lag distribution determines what any window width can recover.
3. **C3. The window-width curve.** For widths of one through twelve years, the share of arcs falling inside a window, the giant-component share, mean degree, and the component-size profile. This is the measurement that turns the choice of era width from a guess into a finding. If most of the available connectivity arrives by three years, the cost of a five-year window is not worth paying.
4. **C4. The whole-corpus graph.** Component profile and degree distribution of the full 590,170-vertex, 1,238,000-arc network. This is the ceiling any windowing can approach, and it is what a single-partition construction would run on.
5. **C5. Directed Leiden at corpus scale.** `delta_modularity_directed_best!` finds a vertex's incoming arcs by scanning the whole adjacency matrix, which costs $O(nnz)$ per vertex. That is tolerable at 50,000 vertices and impossible at 590,000. Either pass a precomputed transpose, as the function's own notes anticipate, or run the corpus undirected. This blocks C4 from becoming a detection run.
6. **C6. Re-derivation under the repaired tie layer.** The descent-survival figures of 0.54% and 1.55%, and the finding that 60.7% of era-23 communities hold exactly one article, were all measured under the pre-repair tie layer and against the legacy partitions. Both need re-running once a construction is settled.

C1 DONE: cited works carried the citing article's year; the rebuild dates them by their own.

C2 DONE: median lag 7 years; backward ties reach a mean of 1.5–1.9 eras. §14.7

C3 PART: a fixed 7-year tiling retains 24.6% of arcs, about half of a sliding window. Giant share still unmeasured.

C4 PART: the full network is now built; its component profile is still unmeasured.

C5: fix the incoming-arc scan, or run undirected, before any corpus-scale detection.

C6: re-derive descent survival and the single-article community share.

7. **C7. The dominance flag.** `champ_community_detection` marks more partitions dominant than ever reach the envelope: era 12 flags 34, while only 19 win at any γ in the swept range [measured]. The crossing test and its 10^{-12} tolerance admit partitions that never become optimal. No conclusion here depends on this, because the elbow rule is applied to a superset, but `n_dominant` should not be used as a diagnostic until it is fixed.

C7: `n_dominant` over-counts; do not use as a

Open: which construction. Three options, with what each costs in era pairs and answerable questions.

Substantive decision 5: Which construction the community stage uses

This is the question that blocks everything downstream, and it should be settled with C1–C4 in hand rather than in the abstract. Three options are on the table.

(a) Wider eras. Aggregating to multi-year windows restores every arc whose endpoints fall within the window, and structure will appear. The cost is era pairs. The analytic window spans twelve years, so non-overlapping five-year windows yield two eras and one consecutive pair, where the present design rests on eleven. RQ4 and RQ11 estimate drift across pairs, and Model II's varying effects are indexed by pair; one pair supports none of that. Sliding windows would give seven pairs but would make eras overlap, which breaks the disjointness the descent relation depends on and moves the construction toward Path B.

(b) One partition over the whole range. Detect communities once on the full corpus network, then treat eras as slices of a fixed community structure. This removes the resolution problem entirely: one partition, one γ, nothing to pool and no scale screen. What it gives up is time-varying community structure. Communities become fixed objects, which makes the trait questions easier and changes the character of anything about communities forming, splitting, or dissolving. Path B's persistence questions largely dissolve with it. It also requires C5 first.

(c) One-year eras, with detection on a coupling network. Keep the present era definition, but build the network that detection runs on from shared references rather than direct citation: two same-year articles are linked when they cite the same work, with the shared works drawn from the full edge list rather than only that era. This keeps all eleven pairs and uses every arc. The difficulty is that only articles receive communities this way, while descent requires cited works to belong to one; assigning each cited work to the community of the article that first cited it would approximate what the current construction already does implicitly.

The choice is not purely technical. Each option changes which of the research questions in §4 the design can still answer, and that comparison should be made explicitly before any code is written.

On hold until a construction is chosen.

Substantive decision 6: What the scale screen is, and what failing it means

This decision is held in abeyance and becomes live again only if a construction is adopted whose networks have structure to screen. Three questions then return, and they constrain one another.

What is screened on. Giant-component share is the natural structural proxy, but the range of modularity across a sweep measures the quantity of interest directly. A threshold specified in advance is preferable to one chosen after observing which eras it excludes. If the threshold must instead be empirical, the document should say so explicitly.

What the sweep runs on. The whole disconnected graph, as at present, or the giant component alone. These need not agree, because the null model's mass is computed over whatever graph is supplied.

What happens to an era that fails. Assign it a γ^* with a warning, exclude it from the size-invariance analysis while still partitioning it at the pooled resolution, or fall back to a fixed value. The second appears most defensible, but it should be an explicit methodological choice rather than a default.

14.7 The Rebuilt Construction, and What Lies Outside an Era

The construction has been rebuilt on one canonical key space and on publication-year dating, and the defect the previous subsection describes is repaired: works now appear in both roles, era networks contain article-to-article citation, and mean degree sits near three rather than two. What a seven-year window still leaves outside is large — three quarters of the citations an era emits — but it is not what we assumed. It is mostly recent, thinly cited work one era away, not citation to canon. Since descent is built from the edge list rather than from the era networks, those ties are not lost to the analysis; what the window shapes is the definition of a community, not the relations between communities.

What was wrong, and what fixed it. The era networks keyed citing articles on their DOI and cited references on the raw Web of Science CR string. Those two identifier spaces never intersect, so a caffeine article cited by another caffeine article became two vertices: no vertex carried both in- and out-degree, and no era network contained a path of length two [measured]. The 2023 pipeline had not made that mistake — it resolved both roles into one key space, a normalized DOI or an `author_year_journal` triple — and the rebuild restores that step. Separately, cited works now carry their own publication year, parsed from the CR string, rather than the year of the article citing them.

The corpus after the rebuild. 589,784 works and 1,238,288 distinct citations [measured], with every node id preserved from `citation_node_list_3Oct2024.Rda`, so the legacy partitions and the known-answer fixtures remain joinable. The 386-work difference from the old list is the merging of labels that differed only in the case of a DOI. Of those works, 20,438 appear both as a citing article and as a cited reference [measured] — 58.2% of the 35,103 caffeine articles are themselves cited by another caffeine article.

The era networks. Seven-year eras, each work placed by its own publication year, an arc kept only when both endpoints fall inside the window.

Era	Years	Vertices	Arcs	Citing	Both degrees	Art.–art. arcs	Mean degree
11	1971–77	2,629	3,889	518	179	309	2.96
12	1978–84	4,268	6,361	897	297	348	2.98
13	1985–91	10,479	15,637	1,627	484	523	2.98
14	1992–98	27,380	44,575	4,244	1,345	2,039	3.26
15	1999–2005	32,848	49,263	4,626	1,676	2,681	3.00
16	2006–12	45,508	67,081	5,891	2,516	4,970	2.95
17	2013–19	76,070	113,008	9,080	4,185	10,794	2.97

All values are measured [measured]. Era 11 is where mean degree reaches its plateau and where the share of citing articles that are themselves cited within the era rises to 35%; by era 17 that share is 46%. Eras 1 to 10 are too small to carry structure — three of them contain no such work at all — and era 18 spans only 2020–2021. The analytic range is therefore eras 11 to 17: seven eras and six consecutive pairs.

For scale against the earlier construction: old era 12, a single year, held 76 vertices with both degrees and old era 23 held 244. New era 17 holds 4,185, about 598 per year, and 10,794 article-to-article arcs where the previous seven-year files had none [measured].

What the window leaves outside. Coverage is narrower than the era tables suggest. The era networks hold 204,042 vertices, 34.6% of the corpus, and 304,602 arcs, 24.6% of the citation record [measured]. A work belongs to one era only, so that is the whole coverage rather than a per-era figure.

The retention figure deserves a note, because it sits against the 52% that the interval analysis reported for a seven-year window. Both are right: a *sliding* window keeps any pair of works published within seven years of each other, while a fixed tiling also requires both to fall on the same side of a boundary. The ratio, 24.6 against 52, is 0.47 — a fixed tiling keeps about half of what a sliding window of the same width would.

What those outside ties are made of. We expected the discarded ties to be citation to foundational work, which a study of ongoing integration has reason to exclude. Three measurements over the 833,949 backward ties emitted by eras 11 to 17 say otherwise [measured].

Era distance	Years	Arcs	Share	Gini over targets	Targets cited once
1	7	464,691	55.7%	0.423	31.7%
2	14	212,432	25.5%	0.393	29.9%
3	21	84,612	10.2%	0.353	30.2%
4	28	37,812	4.5%	0.342	30.4%
5	35	17,273	2.1%	0.331	32.6%
≥6	42+	17,129	2.1%	—	—

Distance. Backward ties are near, not far: 55.7% reach exactly one era back and 81.2% fall within two. Only 18.8% reach three eras or more, which is where citation to canon would live.

Concentration. Canonical citation should land repeatedly on a few works, so concentration should rise with distance. It falls. The Gini over targets drops from 0.423 at distance one to 0.262 at distance nine, and the share of ties landing on a target that receives only one tie at that distance rises from 39.6% to 58.5% [measured]. The further back a tie reaches, the more thinly it spreads.

Works cited once. 31.7% of distance-one backward ties land on a work cited exactly once in the entire corpus — about 147,000 arcs [measured]. A work cited once cannot be canonical, and this measure needs no threshold.

A canonical layer does exist, and it is recognisable: the heaviest targets are Fredholm's 1999 adenosine review with 1,067 backward ties, and then method papers — Lowry 1951 at a mean distance of 6.4 eras across 7 citing eras, Bradford 1976, Laemmli 1970, Grynkiewicz 1985 [measured]. But it is thin. The heaviest 5% of targets absorb a third of the ties at distance one; the rest of the mass is elsewhere.

Why this matters less than it appears, and where it matters. A distance-one backward tie is a citation from era t to a work belonging to era $t-1$. That is the descent relation of §3, and the tie layer builds it from the citation edge list and the era maps, never from the era networks (§14). The 464,691 arcs at distance one are therefore present in the analysis as its primary signal, not discarded.

What the window shapes is the *definition of a community.* Communities are induced from a within-window slice that over-represents the recent, self-citing core of each period, and every model in this document takes the community as its unit. So the caveat is real but specific: the relations we measure rest on the whole citation record, while the objects between which we measure them are built from a quarter of it.

One trend to carry into interpretation. The distance-one share of backward ties falls steadily across the window — 69.3% in era 11 to 51.4% in era 17 — while the share of those ties landing on works cited once rises from 28.9% to 36.0% [measured]. As the literature accumulates, each period reaches further back and onto thinner material. Two consequences for reading the models. A trend in any community-level quantity across eras is partly a trend in what the window captures, so era effects and drift terms should be interpreted against this. And the cross-era share, which the construction now records per work as `pct_out_cross_era`, is a genuine covariate rather than a diagnostic: it measures how far outside its own period a work reaches, and it varies systematically with era.

Substantive decision 7: Whether the window should be fixed, sliding, or supplemented

The measurements above establish what a fixed seven-year tiling costs: roughly half the ties a sliding window of the same width would keep, and the loss falls mostly on recent, thinly cited work one era away rather than on canon. Three responses are open, and the choice should be made before the communities are fixed rather than after.

Accept the tiling. Eras stay disjoint, descent keeps its current meaning, and the bias is documented and carried as a covariate. This is the current plan.

Overlapping windows. More of the near ties enter the community definition, at the cost of disjointness — which the descent relation depends on, and which moves the construction toward Path B.

Lean on the full network. The whole-corpus network is now built alongside the era networks, over the same node ids. Communities detected there would rest on the entire citation record, at the cost of time-varying community structure.

14.8 Model Validation and Predictive Checks

The tests above check that the pipeline builds the right table. The checks below check that the model built on that table is worth believing. Both belong in the pipeline, because both have to run automatically: a check that is performed once by hand and then forgotten is not a check.

The organizing idea is that a Bayesian model is a generative story – it says how the data could have come about – and a generative story can be run forward. If we run it forward and it produces data that look nothing like the caffeine literature, the story is wrong, whatever its coefficients say. Each of the checks below is a version of running the model forward and looking at what comes out.

Prior predictive checks, run before the data are seen. Every parameter in Models I and II carries a prior, and priors on the log or logit scale are notoriously easy to set badly – a prior that looks modest as a number can imply absurd counts once exponentiated. So before fitting, we draw parameters from the priors alone, generate citation counts from them, and ask whether those counts are of a plausible order. Concretely: do the simulated counts stay within a few orders of magnitude of what a community emitting two thousand citations could produce, or does the prior routinely generate a community citing one work ten million times? Do the simulated hurdle probabilities span the full range rather than piling up at 0 or 1? A prior that fails this is replaced and the replacement documented. This is a test of the specification, and it can be run before the data exist.

Posterior predictive checks, run after fitting. Having fit the model, we draw new datasets from the posterior and compare them to the observed one. The comparison is on features we care

about rather than on fit indices:

- the proportion of zeros, which the hurdle stage exists to get right;
- the full distribution of nonzero counts, particularly the upper tail, since a few works are cited very heavily and a model that cannot generate them is misspecified;
- the distribution of community out-degree and in-degree in the reconstructed network;
- the concentration of ties, the HHI of each community's outgoing weight, which is what the tie weightings of §6.2 are built to describe.

Stratified posterior predictive checks. An aggregate check can pass while the model is badly wrong in the region the argument depends on. Since our claims are about particular kinds of pair, we repeat the checks within strata: by topical proximity (does the model fit high-proximity pairs as well as low-proximity ones?), by community size, by cognitive orientation, by literature, and by era. The cognitive stratum matters most. If the model systematically over-predicts citation for cognitively oriented pairs, then every cognitively oriented pair will appear to have a shortfall, and RQ10's list will be an artifact of misfit rather than a finding. This check is the one that stands between the forgone-connection analysis and a fabricated result, and it should be reported whether or not it is flattering.

Leave-one-out cross-validation. Several modeling choices in this document are genuine alternatives rather than settled facts, and LOO – estimated by Pareto-smoothed importance sampling, so that it does not require refitting – gives a common scale on which to compare them by how well each predicts observations it did not see. The comparisons worth running are those where a scientific claim rides on the choice:

- hurdle against a plain negative binomial, testing whether the two-stage structure of §9 earns its complexity;
- with and without the work-level salience effect u_h, testing whether individual works really do differ in the way §7 assumes;
- cognitive orientation by subject matter against orientation by disciplinary provenance, which is a substantive question about which measure captures the phenomenon and not merely a robustness check;
- with and without the coupling term, and with coupling entered linearly against entered as a spline, testing whether the relationship in RQ9 is monotone;
- literature-level varying intercepts against none, testing whether the blocks of §11 explain anything the covariates do not.

Two cautions. Any observation with a Pareto $\hat{k}$ above 0.7 is refit exactly rather than approximated, and the count of such observations is reported – with heavy-tailed citation counts there will be some. And LOO here is ordinary leave-one-out, which asks how well the model predicts a held-out *row*, not a held-out era; it is not a test of forecasting into a future era, and we do not report it as one.

Sensitivity analysis. The forgone-connection list depends on four stated constants: Δ, the cutoff on F, and the two quantile cuts on K and S (§10). We report how the list changes as each varies over a stated range, including which pairs enter and leave, and we report the same for the choice between the two cognitive measures. A result that survives only one setting of these constants is reported as such.

What the pipeline must do with all of this. Each check above is a function that returns a pass, a fail, or a number, and the fitting stage writes them to a report alongside the posterior. The prior predictive check runs before fitting and blocks it on failure. The posterior and stratified checks and the LOO comparisons run after fitting and are recorded whether or not they pass, so that a degradation between one fit and the next is visible in the diff rather than discovered later.